Price Watchdog: Autonomous Multi-Platform E-commerce Agent

Project Domain / Category

Artificial Intelligence / Web Programming / Web Automation

Abstract / Introduction

The **Price Watchdog** is an intelligent AI agent designed to solve the problem of manual price tracking in the highly volatile e-commerce market of Pakistan. Unlike standard scraping scripts that break when a website layout changes, this project utilizes a "Reasoning Agent" that understands web content semantically. The agent can visit a URL, distinguish the main product's price from "Related Products" or "Advertisements," and compare it against a user-defined target.

The project follows a **ReAct (Reasoning + Acting)** architecture. The student will develop a system where a user provides a product link (e.g., from PriceOye or Sapphire) and a desired price. The agent autonomously monitors the site, identifies price drops, and provides a logic-based summary of why a notification was triggered. This project introduces students to the core pillars of AI development: **Prompt Engineering, Tool-Calling, and Agentic Workflows.**

Functional Requirements:

Provide a bulleted list of functional requirements

1. **FR1: URL Processing & Validation** – The system must accept a single product URL and verify its status (Up/Down) before proceeding.
2. **FR2: Intelligent Content Extraction** – The agent must use an LLM to extract the **Product Name**, **Current Price**, and **Currency** from raw web data, specifically ignoring sidebar or "similar" item prices.
3. **FR3: Variant Detection** – If a product has multiple versions (e.g., 128GB vs 256GB), the agent must identify which price it is tracking and log this choice.
4. **FR4: Autonomous Comparison Logic** – The system must compare the extracted price against the user's "Target Price" and determine if an alert is necessary.
5. **FR5: Availability Monitoring** – The agent must detect and report if a product goes "Out of Stock" rather than simply failing to find a price.
6. **FR6: Notification Trigger** – The system must generate a structured alert (Console message or Email) containing the price drop percentage and a direct link to the product.
7. **FR7: History Management** – The system must store previous price checks in a local file (JSON/CSV) to show a simple price trend to the user.

Tools:

1. **Programming Language:** Python 3.10 or higher.
2. **LLM API (The Brain): Google Gemini 2.5/3.0 Flash** (Free Tier via Google AI Studio).
3. **Agent Framework: CrewAI** (Open-source library for defining Agent roles and tasks).
4. **Scraping Tool: Firecrawl** (Free tier for turning websites into LLM-ready Markdown) or **BeautifulSoup4**.

5. **Database: SQLite** (Built-in Python database for tracking price history).
6. **Web Interface (Optional): Streamlit** (Free framework for building the user dashboard).
7. **Development Environment:** VS Code or PyCharm.

Supervisor:

Name: Faizan Tahir
Email ID: fazitahir@vu.edu.pk
Skype ID: faizan.vu@outlook.com